457.557 Water Resources Systems Engineering, Spring 2022

TEST (Closed Book), May-17, 2022 (150 mins, 70 pts total)

Student Number: ____________________ Name: ____________________

I) Multiple Choice Questions (3 pts each, 15 pts total)

1. Water resources systems analysis (WRSA) problems are sometimes referred to "wicked" problems. Wicked problems do not have the following trait:

   (a) Not true or false

   (b) Irreversible decisions

   (c) Clearly defined stopping criteria

   (d) Unknown consequences

   (e) None of the above

2. If A and B are statistically independent, then the following holds true:

   (a) P(A|B) = P(A|B)P(B)

   (b) P(A∩B) = P(A)P(B)

   (c) P(A∩B) = 0

   (d) Both (a) and (b)

   (e) All of the above

3. Choose the correct word(s) for the blank in the following sentence:
   Among various measures of the central tendencies, _________ can be classified as a reduced measure of central tendency that is less sensitive to outliers.

   (a) the median

   (b) the mode

   (c) correlation coefficient

   (d) arithmetic mean

   (e) both (a) and (b)

4. Choose the correct phrase(s) for the blank in the following sentence:
   A Pareto front ___________________.

   (a) must be convex, differentiable, and continuous.

   (b) requires linear models.

   (c) is probabilistic, with unknown values.

   (d) contains only non-dominated points.

   (e) all of the above

5. Choose the correct phrase(s) for the blank in the following sentence:
   An ideal solution vector in a multi-objective problem ______________.

   (a) must be dominated by each Pareto optimal set.

   (b) must lie on the Pareto front.

   (c) must dominate any vector in the feasible solution space.

   (d) is always infeasible

   (e) all of the above

II) Short Answers (7 pts each, 35 pts total)

6. Your industry should choose one of the following two goods for sale. Based only on the information provided below, which good should be fabricated?
   (hint: $PV(x, n, r) = x/(1+r)^n$)

   A. Good A: costs $1,000 paid up-front to fabricate, takes 6 months to fabricate and be ready for sale, and would sell 6 months from now for $2,000 in a market with 2% discount rate per month.

   B. Good B: costs $1,500 paid up-front to fabricate, takes 7 months to fabricate and be ready for sale, and would sell 7 months from now for $2,750 in a market with 3% discount rate per month.

7. Solve the following problem and provide answers to the following questions:
   (a) Solve the following problem with the Lagrange multiplier method:
   (i.e. What are the optimal values of x_1^*, x_2^* and λ?)

$$\text{Minimize } (x_1-1)^2+(x_2-2)^2$$

$$\text{Subject to: } x_1-2x_2=0$$

   (b) What is the meaning of the Lagrange multiplier? If a what can be inferred from the fact that a multiplier for the first constraint (λ_1) is greater than the multiplier for the second constraint (λ_2)? (i.e. $\lambda_1>\lambda_2$)
   (c) What are the possible limitations arising from using the calculus-based optimization algorithm such as hill climbing & Lagrange multiplier?

8. You have a box with two coins. One of them is red, and the other blue. We have P(head|red)=1/4 and P(head|blue)=1/2. You pick a coin uniformly at random and toss it. It comes up heads. What is the probability that it was a red coin?

9. The table below gives values of two objectives from a multi-objective maximization problem:

F1	F2
10	1
8	3
6	11
4	8
2	4
0	2

   (a) Identify the ideal point in the following problem.
   (b) Which points are on the Pareto front?

10. Name two types of ANN models and briefly explain the difference between the two. What is the key difference between the two types of ANN models?

III) Problem Solving/Long Answers (10 pts each, 20 pts total)

11. Refer to the following recursive equation of stochastic dynamic programming, and provide answers to the following questions:

$$f_t(S_t, Q_t) = \max_{R_t}\left[B_t(S_t, Q_t, R_t) + E_{Q_{t-1}|Q_t}[f_{t+1}(S_{t+1}, Q_{t+1})]\right]$$

(a) What are the three curses of SDP mentioned in class? Name the three curses and briefly explain about them using the recursive equation of SDP provided above.

(b) Using the same notations from the above recursive equation, write the recursive equation for deterministic DP. Then, briefly explain the steps you need to take to solve the deterministic DP problem. You may include figures/graphs during your explanation if necessary.

12. The shaded region is the feasible solution space for a multi-objective optimization problem.

(a) Highlight the Pareto front and identify the ideal points for both plots (a) and (b). The objectives of two different problems are as the following:

Plot (a): minimize f_1 & minimize f_2

Plot (b): maximize f_1 & maximize f_2

(b) What is the concept behind Pareto optimality? What is the key difference between Pareto optimality and the Bellman optimality that DP is based on?